設計過程中認知層的交互連結
The interlinked design process

唐玄輝 HSIEN-HUI TANG

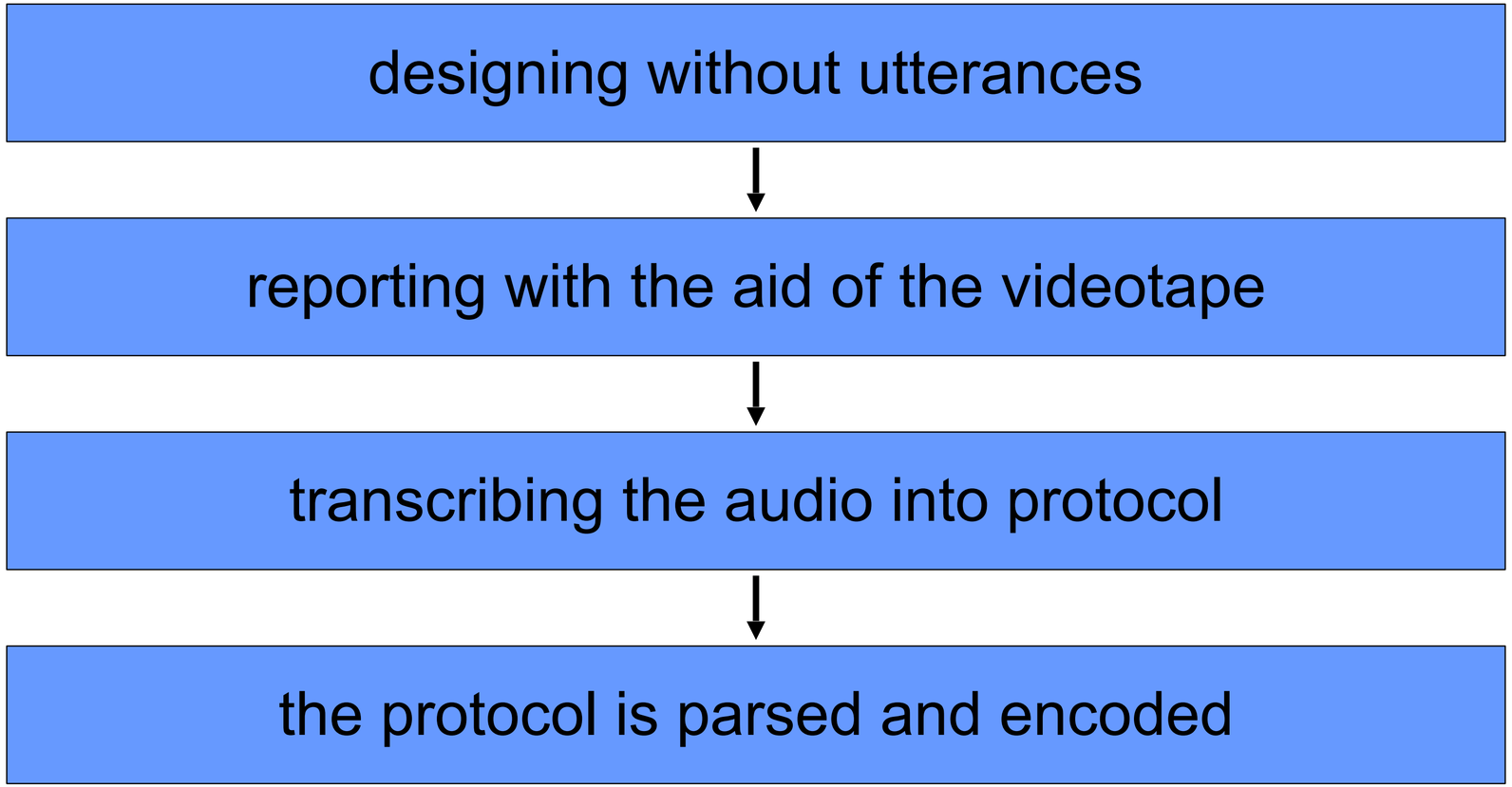
Graduate Institute of Architecture

National Chiao Tung University Taiwan

Conceptual Designing

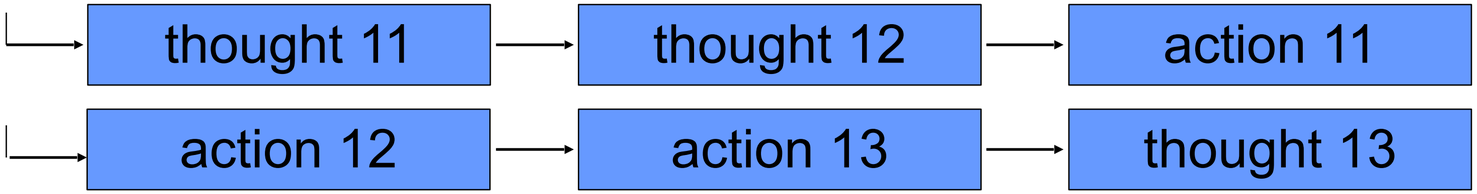
- Conceptual designing occurs when the designer initially tries to understand the design problem and sets the situation for the following processes.
- Conceptual designing is the period having the richest range of ideas, problems, and creativity.
- Current CAAD systems seem to be absent from this conceptual designing.
- CAAD systems should have the capacity to support conceptual designing to provide a holistic design process support.

Retrospective Protocol Analysis



The Definition of Segments 段句

- one *segment* accounts for a designer's single intention, and therefore consists of pieces of information that appear to have occurred simultaneously in the designer's mind.
- flagged by



- the length

Speed of Intention Shift 意圖移轉

- Segments reflect the designer's intentions.

	Number of segments	Average time span	Duration of experiment
Novice	145	20 sec	48 min
Expert	338	8 sec	45 min

- The time this expert took to shift his intention was not enough for CAAD
- Why designers still prefer using pens and papers
- It is simply because these familiar sketching skills can catch up with the speed of thought, the speed of ideas, and even the speed of creativity.

Design Content-Oriented Coding Coding Scheme

- Established by Suwa, Purcell, and Gero (1998, 2000)
- Applied in a series of research (Gero & Tang, 1999) (Tang & Gero, 2000, 2001a, 2001b, 2001c)
- Understand the cognitive aspects of the design process
- Consists of four cognitive levels that imply the sequential process a stimulus goes from the external world to the internal world
- DCOCS (pronounced as dee-cokes)

Levels in DCOCS Scheme 基模

PHYSICAL: the instances that have direct relevance to the external world

↓ ↑

PERCEPTUAL: the instances of attending to visuo-spatial features/relationships in a automatic perceptual mechanism

↓ ↑

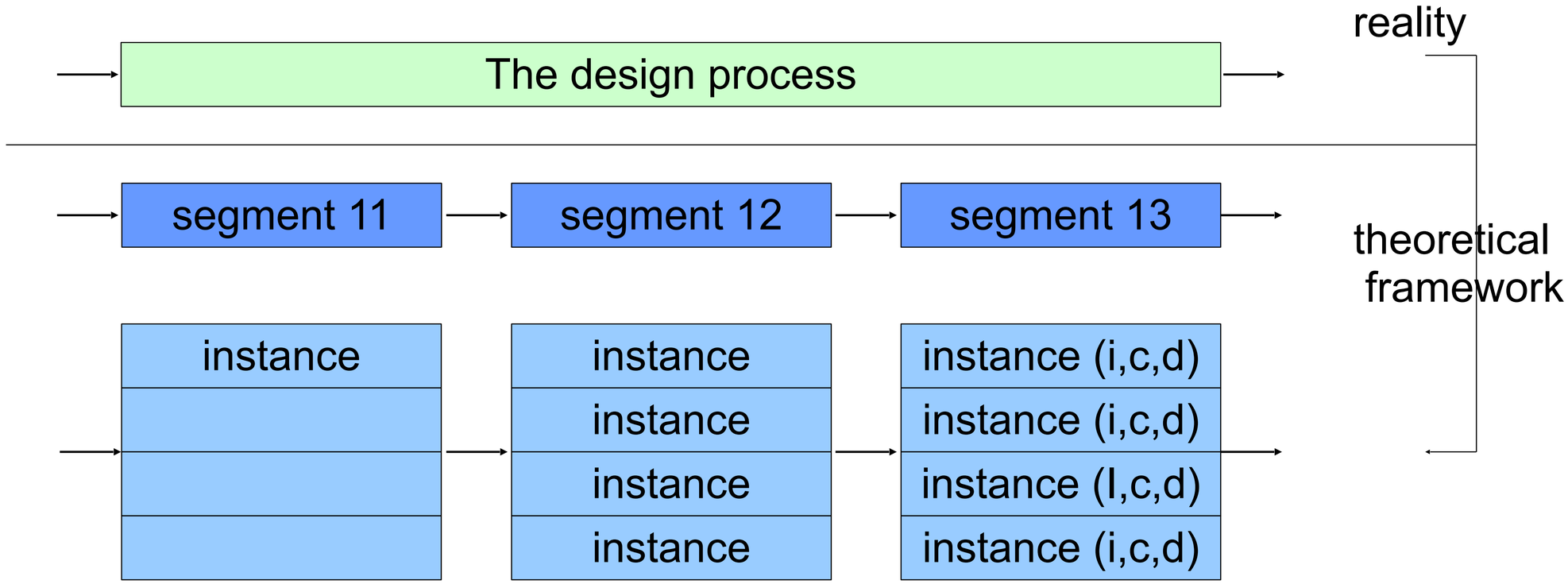
FUNCTIONAL: the functional references mapped between visuo-spatial relationships and meanings

↓ ↑

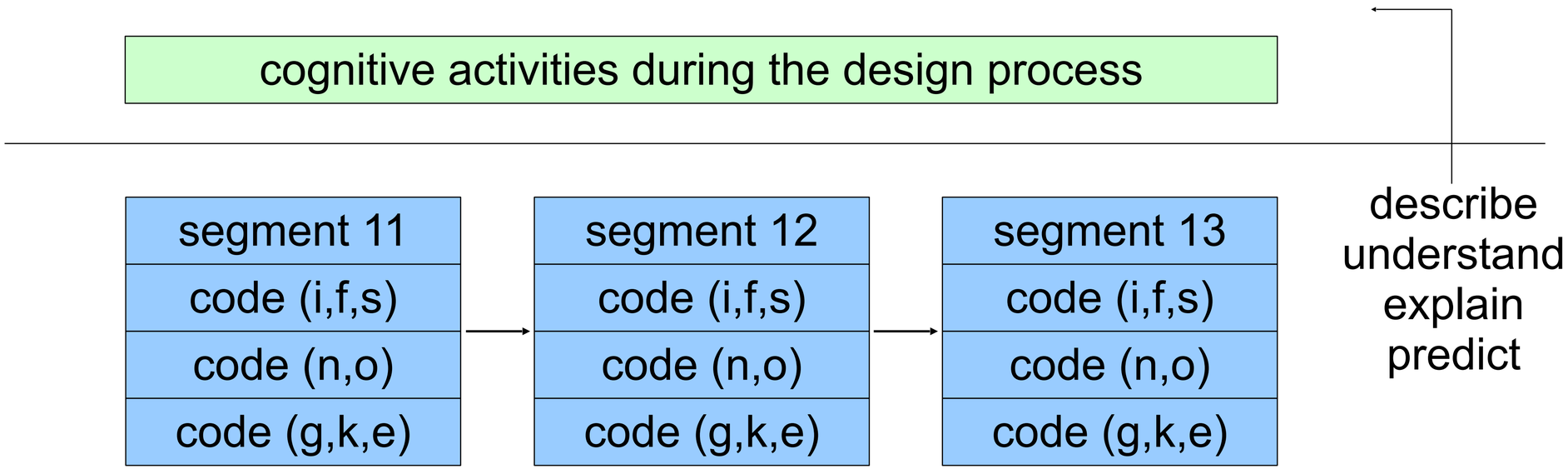
CONCEPTUAL: the instances that process abstract concepts and physical and perceptual levels

Coding Scheme 編碼基模

- a coding scheme categorizes every single segment into different levels.



A Representational System



Justification for DCOCS

- Applications
  - externally, activities highly related to the role of sketches
  - internally, functional references, knowledge retrieval, and aesthetic evaluation
- Validity and reliability have been proved
- Advantages
  - links the knowledge between cognitive psychology and design disciplines
  - able to repeat the experiments and analyses to verify our findings.

Instances in DCOCS 單一活動案例

- The instances in DCOCS refer to observed actions of a specific activity in a level; for example, depicting a line is an instance in the physical level, so it is a physical action.
- Three kinds of physical instances: drawing, looking, and others.
- There is only one kind of instance in perceptual, functional, and conceptual levels.
- An instance of perceiving is a perceptual action, an instance of function-referring is a functional action, and an instance of goal-setting is a conceptual action.
- Each level may have more than one instance, and instances in different levels have interrelated dependencies.

An example

Transcript

37 00:12:13 then so I see, now I am pretty inside the building, this was a cafe, real profit to the building, I thought a cafe might be here, so people can walk here and see the view and enjoy outside, whatever area, so I

D_TYPE		INDEX	BASEON	L_TYPE	L_CONTENT
Dc	1	new		L2	1
Dc	2	new		L1	7
Dwo	1	new	Fn1		9

P_TYPE		INDEX	P_CONTENT	INDEX	BASEON
Prp	1	new	direction: the common space to the	old	L2
Prn	1	new	arrangement: the coffee and the	new/old	L1/Dc1/L2

F_TYPE		INDEX	F_CONTENT	INDEX	BASEON
Fn	1	new	coffee	new	Dc1
Fn	2	new	sse the view and enjoy the outside	new	Prp1
Fn	3	new	profit place for the building	new	Dc1
Fo	1	old	building	old	L1
Fn	4	new	walk out from coffee	new	Prn1

G-TYPE		F_CONTENT	TRIGGER
Type1.4	-1	put the coffee in the corner, so that	Fn1, Fn2,Fn4

The Observed Frequencies of Instances

- We calculate the mean and standard division of the observed frequencies of the instances in different levels.

	Novice (20sec)		Expert (8sec)	
	Mean	Std. Dev.	Mean	Std. Dev
Physical instance	3.6	2.3	3.1	1.5
Perceptual instance	1.5	1.2	1.8	1.1
Functional instance	1.9	1.6	2.3	1.6
Conceptual instance	0.8	0.5	0.8	0.8

- Measure when designers used pens and papers
- The richness of information processed during conceptual designing.
- It is hard to image how designers can use CAAD systems in conceptual design

Durations of Instances

- We divide the average time span by the number of instances: the expert's actions alter more frequently than the novice
- We should start to think about the CAAD system which is capable of supporting a designer's perceptual and functional instances

	Novice	Expert	Current CAAD
Physical action	5.6 (sec)	2.6	AutoCAD ArchiCAD
Perceptual action	13	4.4	Cognition-based CAAD
Functional action	10.5	3.5	Cognition-based CAAD
Conceptual action	25	10	Knowledge_based CAAD

Three Modes in Conceptual Design

- We further examine the inter-relationships between physical, perceptual, functional, and conceptual levels
- The design process may be regarded as a collection of modes, showing the directions of thinking process among four cognitive levels.
- A sensor-driven process is triggered by physical actions, such as designers' drawing and seeing.
- A concept-driven process is executed by the conceptual actions such as designers' goal-setting
- A co-evolutionary process is one in which physical, perceptual, functional, and conceptual actions interact with each other

Sensor-driven Mode 感官驅使模組

- 42 percent of the expert's segments and 29 percent of the novice's were pure sensor-driven. No conceptual action was involved in these segments.

Transcript

141 00:48:42 Uhh, I say, this looks like a small gap. Maybe people can go out of there.

D_TYPE		INDEX	BASEON	L_TYPE	L_CONTENT
Dsy	1	new	Fn1	L1	18
				L2	16
				L3	17

P_TYPE		INDEX	P_CONTENT	INDEX	BASEON
Prp	1	new	between cafe, gift shop and the	old/old	L1/L2/L3

F_TYPE		INDEX	F_CONTENT	INDEX	BASEON
Fn	1	new	people can go out	new	Prp1
Fo	1	old	the display area	old	L1
Fo	2	old	the cafe	old	L2

G-TYPE		F_CONTENT	TRIGGER
Type1.4	-1	decide the place to go out	Fn1

Concept-driven Mode 意圖驅使模組

- Our empirical data found no pure concept-driven process.
- A cluster leading by this functional instance establishes the basis for conceptual instances as operators for reasoning.

Transcript

28 00:09:06 I was also thinking of circulation within the site, say, you enter here, going to the exit around here, making two roads, I was thinking to ..

D_TYPE	INDEX	BASEON	L_TYPE	L_CONTENT
			L1	9
			L2	1

P_TYPE		INDEX	P_CONTENT	INDEX	BASEON
Por	1	old	between the inner access road and	old/old	L1/L2

F_TYPE		INDEX	F_CONTENT	INDEX	BASEON
Frei	1	new	circulation within the site	old	Por1
Fnp	1	new	making two roads		

G-TYPE		F_CONTENT	TRIGGER
Type1.1	-1	thinking of the circulation within site	Frei1

Co-evolutionary Mode 共同演化模組

Transcript

37 00:12:13 then so I see, now I am pretty inside the building, this was a cafe, real profit to the building, I thought a cafe might be here, so people can walk here and see the view and enjoy outside, whatever area, so I

D_TYPE		INDEX	BASEON	L_TYPE	L_CONTENT
Dc	1	new		L2	1
Dc	2	new		L1	7
Dwo	1	new	Fn1		9

P_TYPE		INDEX	P_CONTENT	INDEX	BASEON
Prp	1	new	direction: the common space to the	old	L2
Prn	1	new	arrangement: the coffee and the	new/old	L1/Dc1/L2

F_TYPE		INDEX	F_CONTENT	INDEX	BASEON
Fn	1	new	coffee	new	Dc1
Fn	2	new	sse the view and enjoy the outside	new	Prp1
Fn	3	new	profit place for the building	new	Dc1
Fo	1	old	building	old	L1
Fn	4	new	walk out from coffee	new	Prn1

G-TYPE		F_CONTENT	TRIGGER
Type1.4	-1	put the coffee in the corner, so that	Fn1, Fn2,Fn4